

Evaluating Digital Demand Signals and Territorial Mobility Patterns to Support Airline Seat Allocation

International Outbound Tourism Analysis in Spain (2019-2025)

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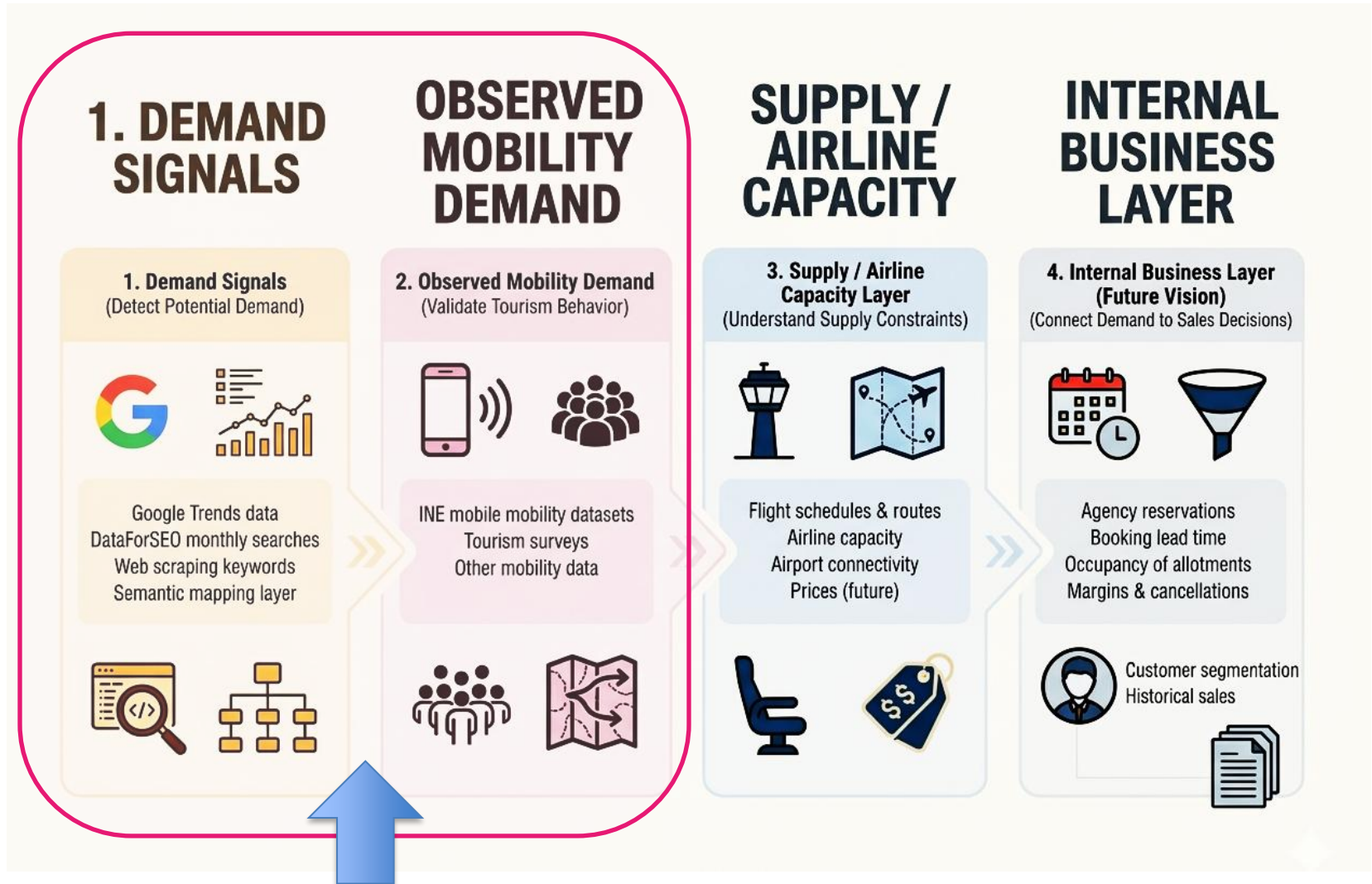
THE BUSINESS PROBLEM

Contractual and Financial Risks of Tour operators and Travel Agencies:

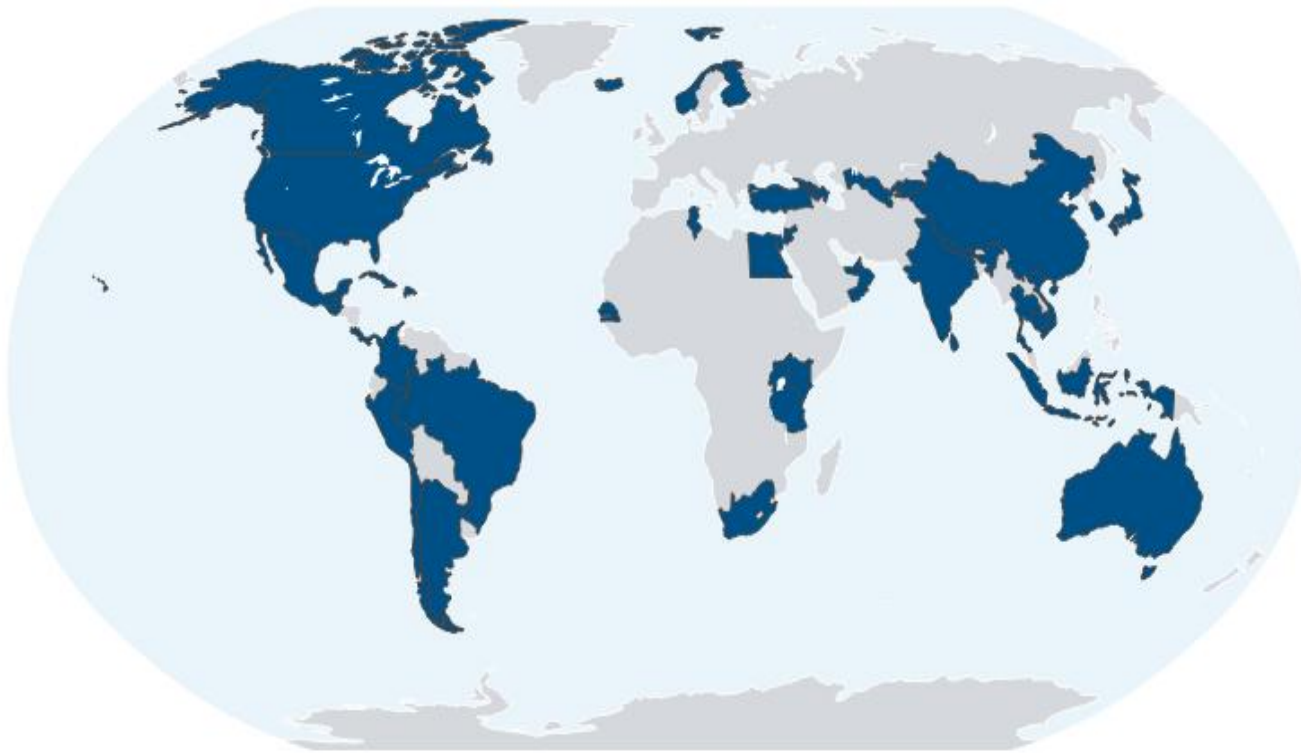
- Flight allotments decided 11-12 months in advance.
- They imply high deposits and other contractual risks
- Unsold seats create financial losses.

**Can digital demand signals anticipate
outbound tourism demand to support
better airline seat allocation decisions?**

COMPLETE PROJECT FRAMEWORK



ANALYTICAL SCOPE: long-haul, agency-mediated destinations



■ Not in scope — core Europe, border mobility, VFR demand

Why exclude Europe?

Among leisure travelers, **45%** of trips to América use travel packages vs only **18%** for Europe.

The “**viaje a**” search signal aligns with agency-mediated demand, where package rates are **2–2.5x** higher than Europe.

Source: INE ETR 2019–2025

The subset was designed to reduce demand noise and focus on destinations where digital search behavior is more likely to reflect organized leisure travel; 45 countries +USA (excluded in some analysis).

DATA SOURCES



INE Mobility

Experimental statistics on outbound mobility (2019-2025) derived from **mobile phone signaling**.

- ~75% coverage of the Spanish market.
- Granularity: Province-Country-Month.
- Used as the **ground truth** for real demand.

Observed demand



Digital Signals

Analysis of search intent and micro-dynamics via global search telemetry.

- **DataForSEO**: Absolute monthly search volumes (high stability).
- **Google Trends**: Relative search interest indices.
- Mapped through a 12-month **lag analysis**.

**Search interest
(absolute & relative)**



Intelligence Layer

Semantic enrichment and domain curation to transform raw data into operation intelligence.

- **Web Scraping**: Commercial keywords from sector-leading websites.
- **Semantic Mapping**: Curation of multi-country regions (Balkans, Indochina).
- **Geospatial Data**: CCAA and Province integration.

Keyword generation

METHODOLOGICAL FRAMEWORK

Two alternative explanatory signals are compared:

- ``trend_index``
 - relative normalized Google Trends intensity (0-100 scale)
- ``monthly_searches``
 - estimated absolute monthly search volume

A strategic subset of destinations is used in order to reduce structural noise from markets dominated by non-tourism travel behavior, including:

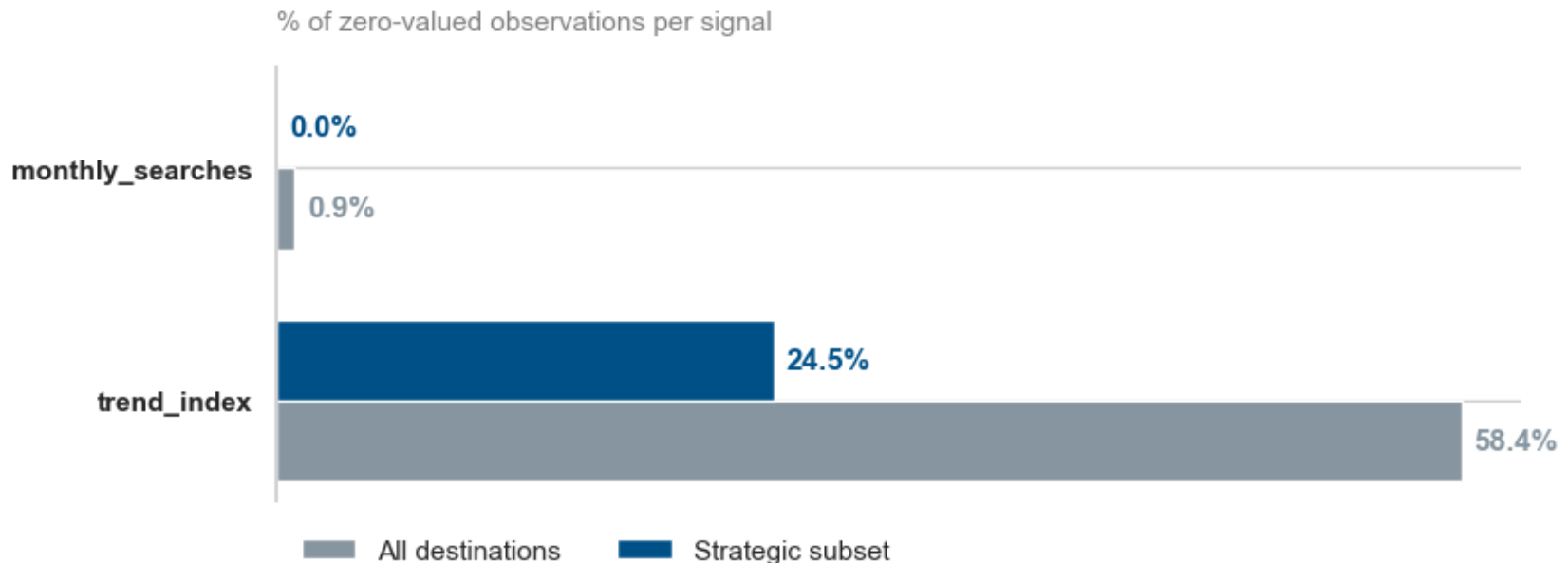
- cross-border commuting,
- family visits,
- migration-related mobility,
- business-heavy destinations.

The goal is not to force predictive performance, but to evaluate where digital demand signals appear useful, unstable, or methodologically limited.

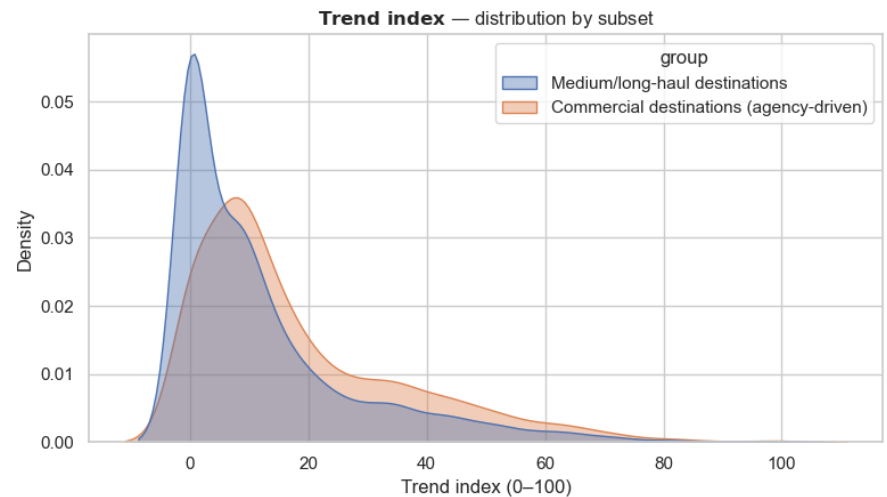
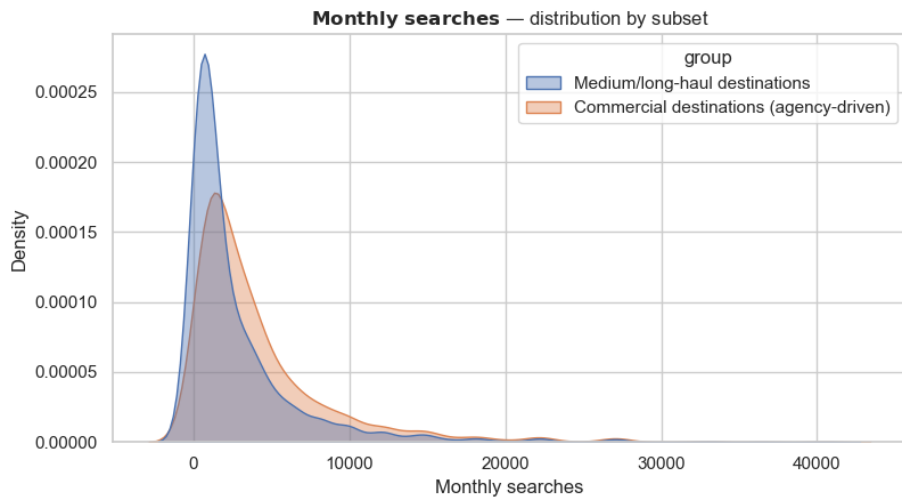
MONTHLY SEARCHES vs GOOGLE TRENDS

Signal quality: % of zero observations

Filtering for strategic destinations reduces months with missing demand signals



SIGNAL DISTRIBUTIONS: Strategic filtering improves data quality while preserving market structure



- **Market Alignment between subsets**
- **Noise Reduction:** filter drops zero-search countries.
- **Demand distribution:** concentrated in a few top destinations (long tail).

NORMALITY TEST JUSTIFIES SPEARMAN

- Shapiro-Wilk Test:

H0: the sample has a Gaussian distribution.

H1: the sample does not have a Gaussian distribution.

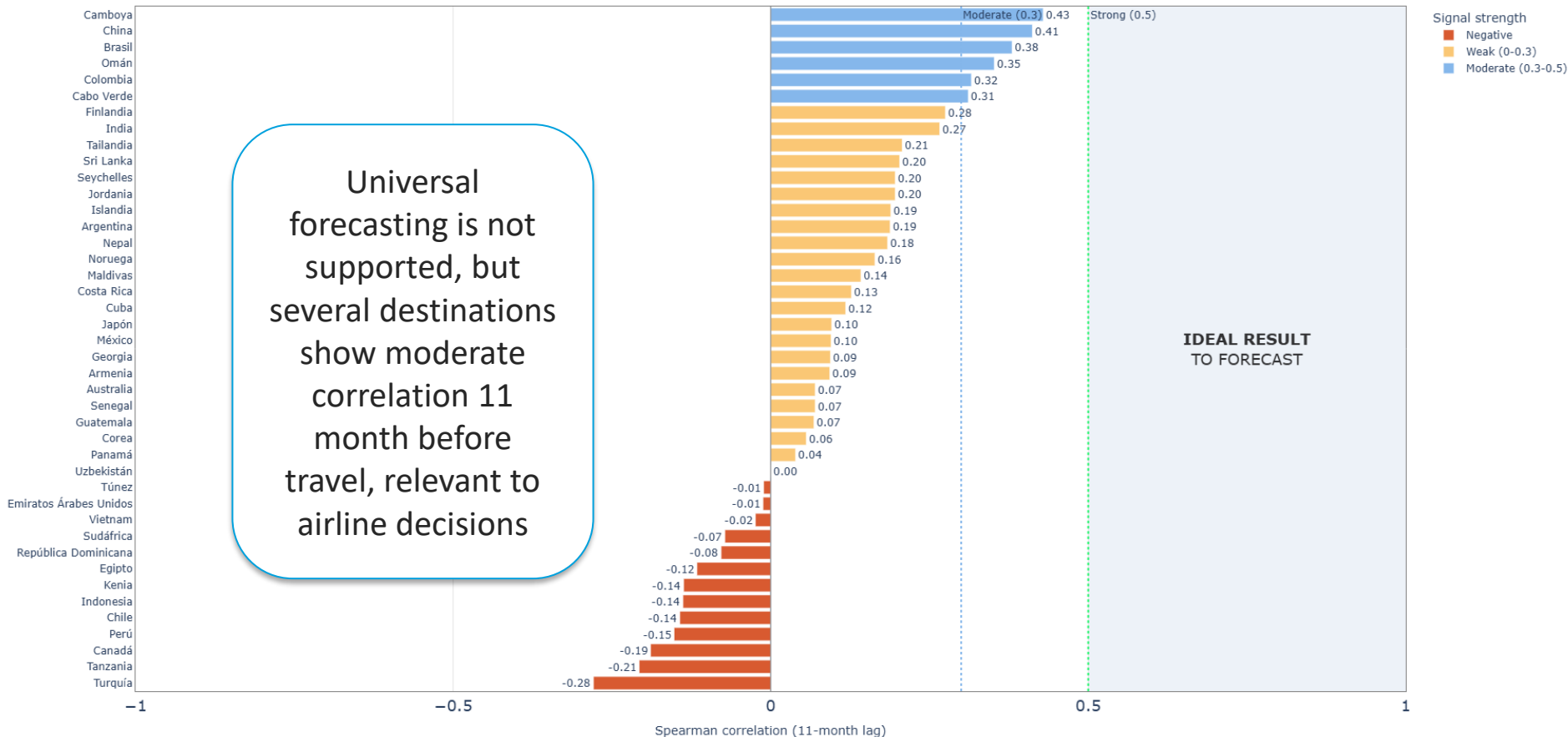
Variable	Shapiro-Wilk W	p-value	Result
total_tourists	0.332	≈ 0	Not normal
monthly_searches	0.711	≈ 0	Not normal
trend_index	0.857	≈ 0	Not normal

$p < 0,05$, H1 aceptada

CORRELATION RESULTS (Lag 11)

Google searches ("viajar a "+destination) vs total trips

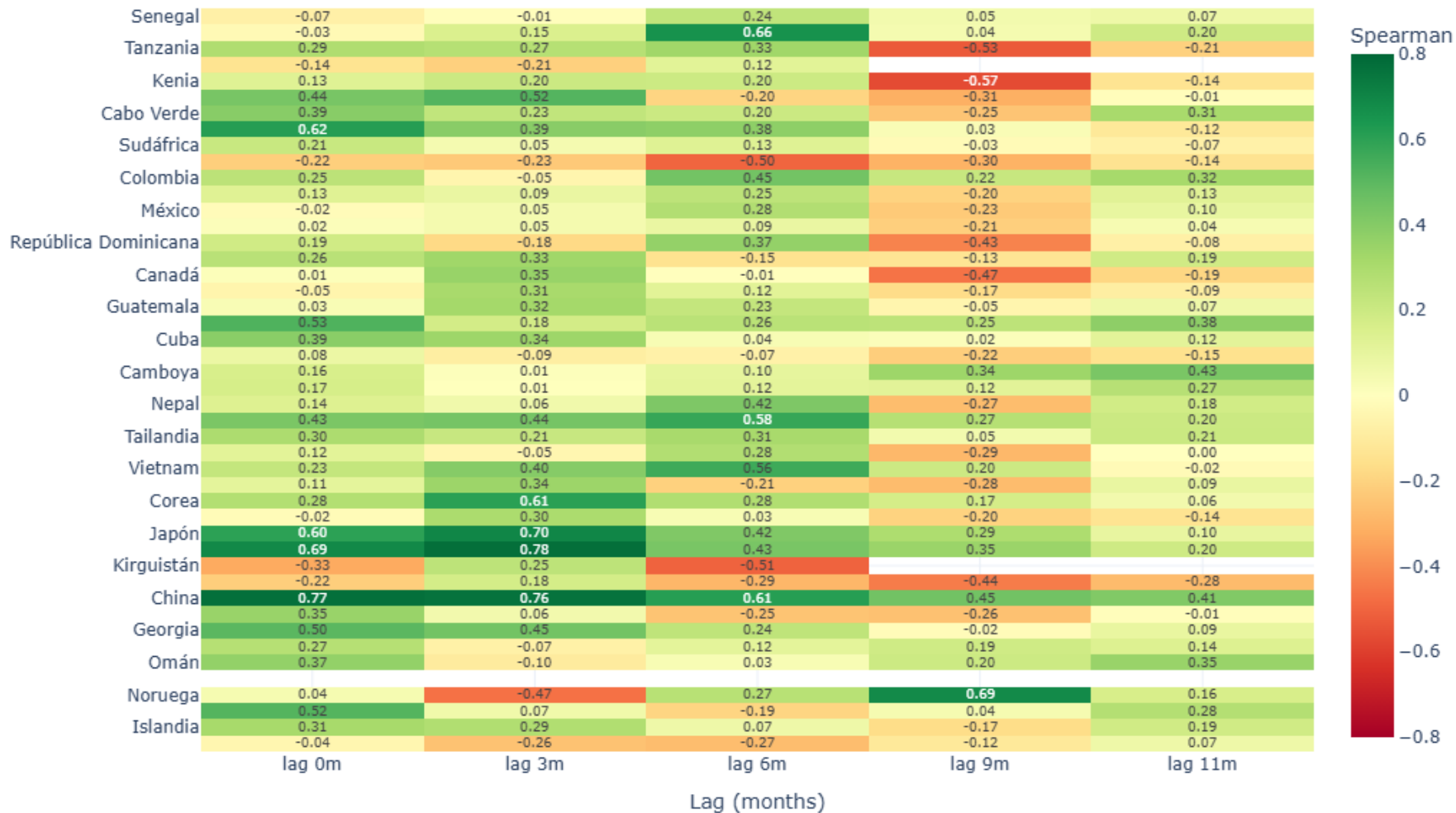
Low correlation at lag 11 across 45 strategic destinations



6 of 45 destinations (13%) showed useful signal at airline planning horizon

CORRELATION RESULTS (different lags)

Search-demand correlation at strategic lag windows (excl. COVID 2020-2021)



WHY CORRELATIONS REMAINED LIMITED?



Mobile-phone measurement dataset limitations



INE data not segmented by travel motivations



Simplified keyword architecture ("viaje a + destination"), not capturing all search behavior.



Searches were only analyzed in Spanish



Not all travellers search online

TOURISM INTELLIGENCE APP - Streamlit

Useful output to explore related business questions and personalize destination focus analysis.



<https://spain-outbound-tourism-analysis-2026.streamlit.app/>

Other business oriented questions (not in app)

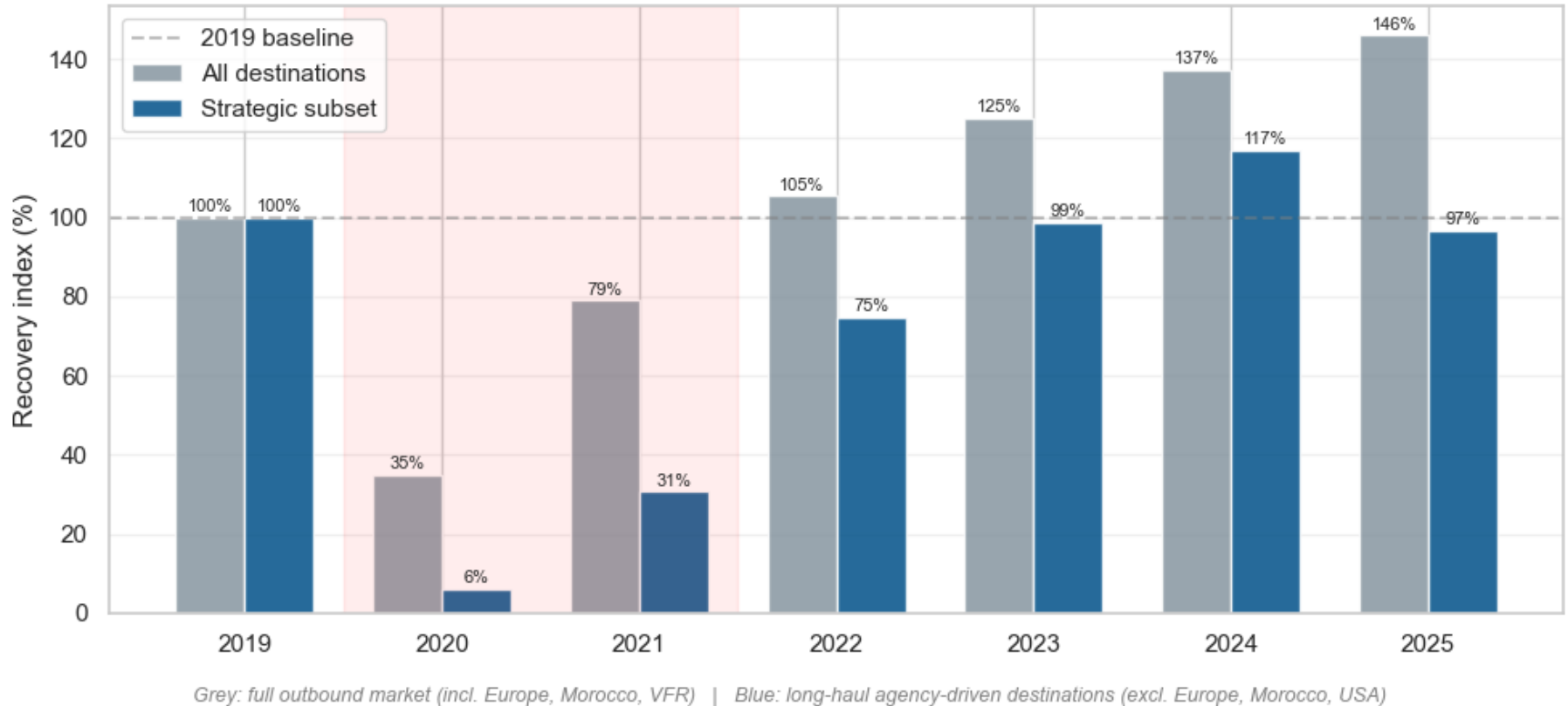
Has the agency aligned with trips recovery after COVID?

If we want to expand besides Madrid and Barcelona, which other provinces generate the highest international outbound demand?

If a destination is successful, what are additional top demand months for each destination, besides August?

COVID-19 impact and recovery dynamics (jul-dec)

Long-haul travel collapsed more and recovered slower

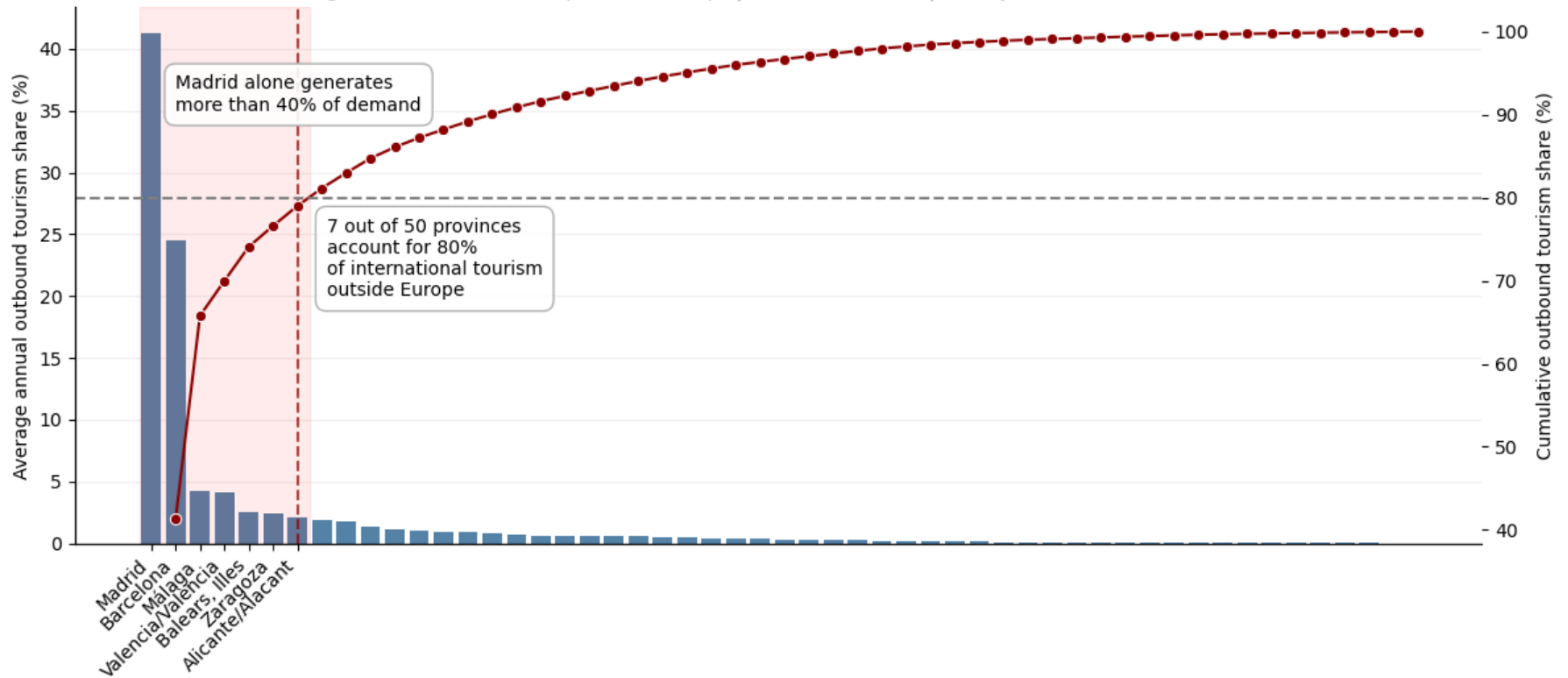


H2 recovery index by segment (Jul-Dec 2019 = 100%)

CONCENTRATION OF OUTBOUND TOURISM

Spain's outbound tourism outside Europe is highly centralized

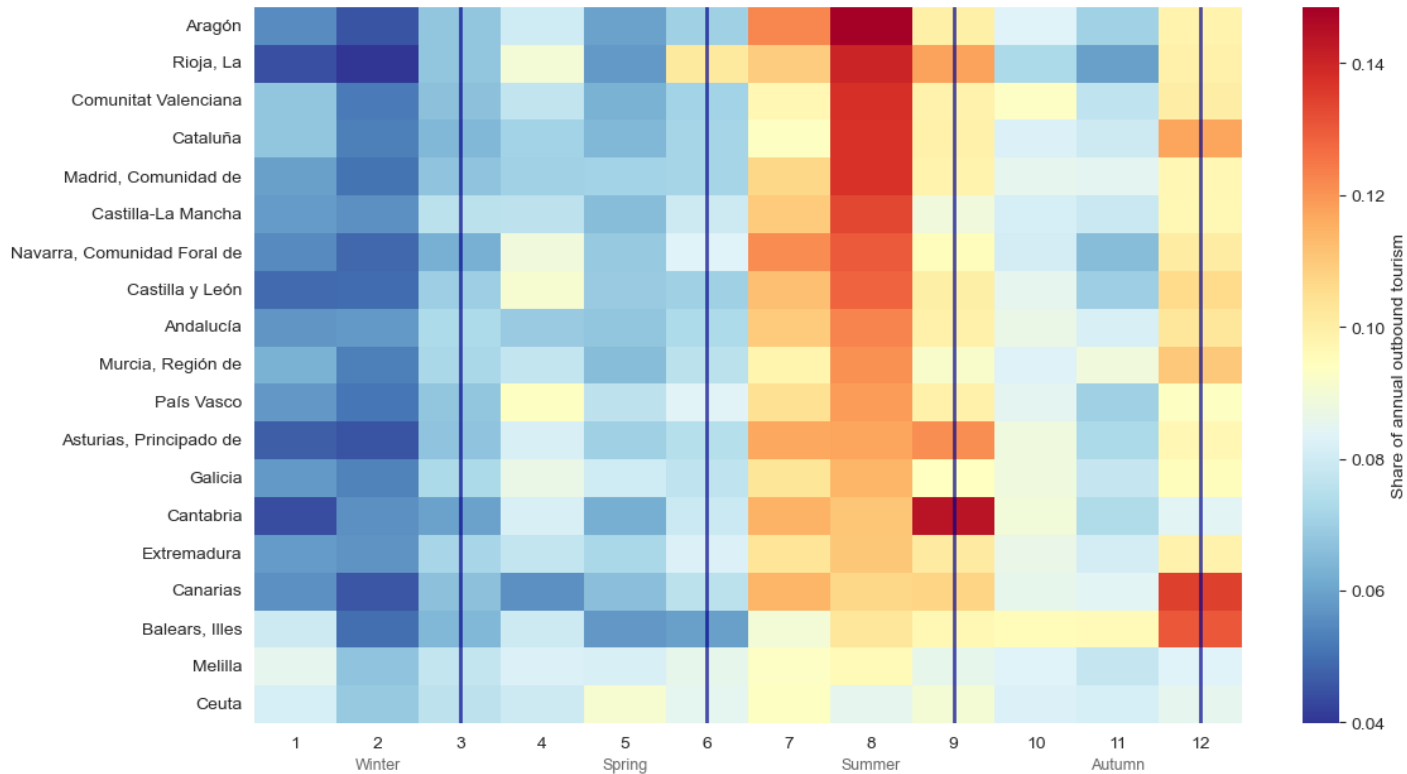
Average annual international trips outside Europe | 2019 & 2022-2025 | Norway, Finland and Iceland included



SEASONALITY PATTERNS

Spanish outbound tourism seasonality varies substantially across regions

Outbound tourism seasonality by autonomous community



Interior regions exhibit strong summer dependence

Metropolitan regions show flatter seasonality profiles

Island territories display stronger winter activity

CONCLUSIONS

Can digital demand signals anticipate outbound tourism demand to support better airline seat allocation?

→ **Partially.** Useful for 13% of destinations at annual horizon



- Search signals show selective predictive value



- Google searches outperforms Google Trends.



- Signals work best as real-time monitors (lag 0–3 months)



- Search signals are a **complement**, not a substitute, for INE mobility data in seat allocation decisions.

NEXT DEVELOPMENT STAGES

From digital demand signals to operational tourism intelligence



Improve signal quality

- Data sources with reason segmentation
- Search-query design
- Additional explanatory variables



Build operational forecasting models

- Autoregressive features
- Macroeconomic indicators
- Transport connectivity



Improve agency intelligence tools

- Interactive dashboards
- Airline capacity integration
- Commercial KPIs

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